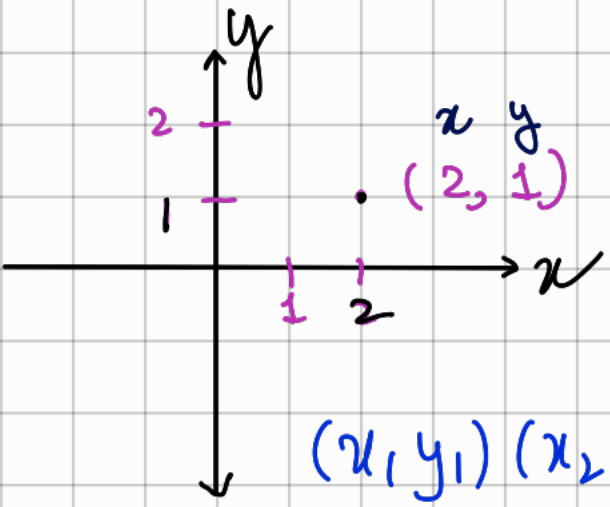
Coordinate Geometry

– Notes by Ishani



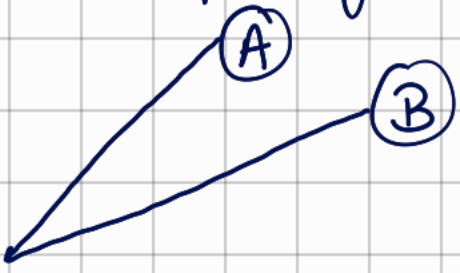
(x, y)

(x_1, y_1) (x_2, y_2)

Midpt: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Gradient / Slope $= \frac{y_2 - y_1}{x_2 - x_1}$

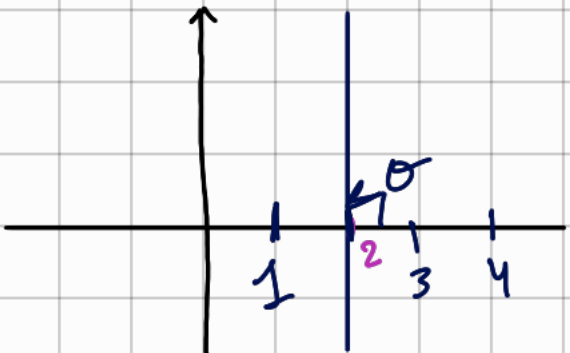
What is Gradient?
= Steepness of line



Which is more steep?
(A)

$\therefore$ gradient of A > gradient of B

(1) Vertical line –



$x = 2$

Gradient = ∞

(x, y)

Eg (x_1, y_1) $(4, 3)$ (x_2, y_2) $(6, 5)$

$\left(\frac{4+6}{2}, \frac{3+5}{2}\right)$

$= \frac{10}{2}, \frac{8}{2}$

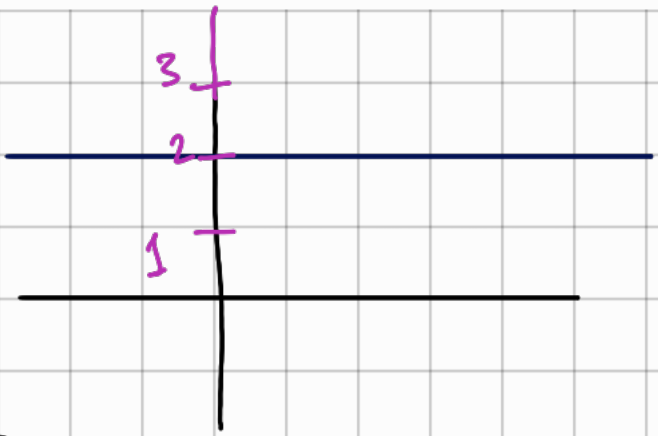
Midpt $= (5, 4)$

gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{5-3}{6-4}$

$= \frac{2}{2}$

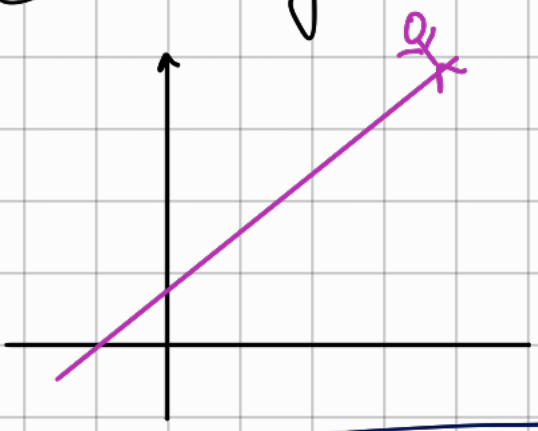
$= 1$



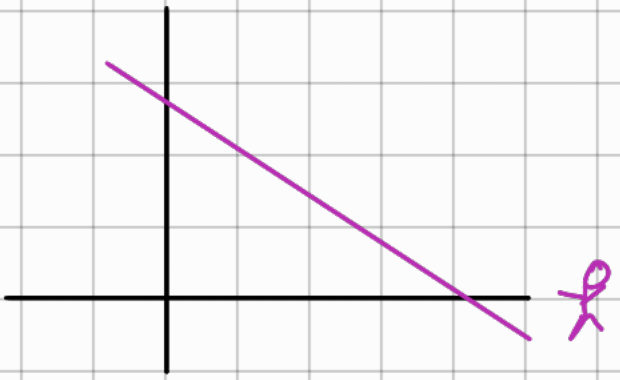
$y = 2$

[gradient = 0]

② Slanting lines –

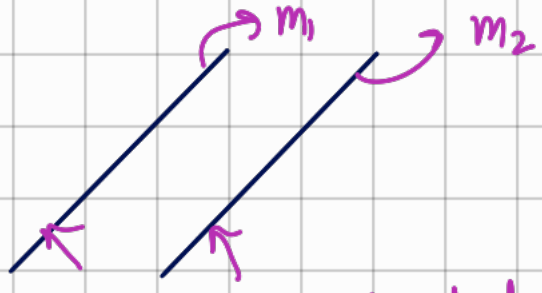


gradient = +ve



gradient = -ve

Parallel lines



→ Two lines if extended never meet

gradient $m_1 = m_2$

→ Same Steepness

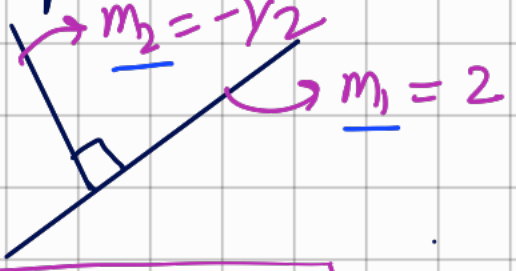
$-(-2)$ 2 → $-\frac{1}{2}$

$5 \longrightarrow -\frac{1}{5}$

$-2 \longrightarrow +\frac{1}{2}$

$-(-2) = +2 \rightarrow \frac{1}{2}$

Perpendicular line



$m_1 m_2 = -1$

negative reciprocal of each other

$5 \rightarrow \frac{1}{5}$

$\frac{1}{4} \rightarrow \frac{-4}{1}$

$m_1 = \frac{-1}{m_2}$

$m_1 m_2 = -1$

$2\ m_2 = -1$

$m_2 = -1/2$

Q line A has gradient $\frac{5}{6}$

line B has gradient $-\frac{6}{5}$

Check if perpendicular

$m_1 \times m_2 = -1$ → multiplication of slope = -1

Yaa, $\frac{5}{6} \times -\frac{6}{5} = -1$

Q2 Given two lines are perpendicular

Slope of one line = 7 → $-\frac{1}{7}$

slope of other line = ?? $-\frac{1}{7}$ ✓

$m_1 = \ominus 1$ / m_2

$7 \times -\frac{1}{7} = -1$

lines are perpendicular

$m_1 \times m_2 = -1$ $m_2 = \frac{-1}{m_1}$

Given lines are perpendicular, find gradient of other line

I	II	
2	$-\frac{1}{2}$	-1
-2	$\frac{1}{2}$	-1
$\frac{5}{8}$	$-\frac{8}{5}$	-1
$-\frac{5}{8}$	$-\left(-\frac{8}{5}\right) = \frac{8}{5}$	-1

Equation of line –

$$y = mx + c$$

y-intercept

gradient

Use this when
1. Gradient is given / slope
2. y intercept (c)

$$y - y_1 = m(x - x_1)$$

Use when
1. Slope (m) / gradient } given
2. Apt on line (x_1, y_1)

Eg Find eq, Slope / gradient = 11 and it cuts y at 7. (y intercept)

$$y = mx + c$$
$$y = 11x + 7$$ ✓

Eg Find eq of line slope = 3 and it passes through $(1, 2)$

x_1 y_1 → pt on line

$$y - y_1 = m(x - x_1)$$
$$y - 2 = 3(x - 1)$$
$$y - 2 = 3x - 3$$
$$\boxed{y = 3x - 1}$$ Ans

Eg Find eq of line slope $= -3$ and it passes through $(5, -2)$

$y = -3x - (-12) = -3x + 12$

$y + 2 = -3(x - 5)$

$y + 2 = -3x + 15$

$y = -3x + 15 - 2$

$y = -3x + 13$

Question → How to find gradient of line from its equation

Eg: $3x + 2y = 7$ ✓ Find gradient

$y = mx + c$

$2y = 7 - 3x$

$y = \frac{7}{2} - \frac{3}{2}x$

gradient $= -\frac{3}{2}$

Eg $x - 2y = 6$

$x - 6 = 2y$

$\frac{x}{2} - \frac{6}{2} = y$

$\frac{1}{2}x - 3 = y$

gradient $= \frac{1}{2}$

Eg $6x - 7y = 17$ Find gradient

$y = \frac{17}{-7} + \frac{6x}{-7} = -\frac{17}{7} + \frac{6}{7}x$

gradient / slope $= +\frac{6}{7}$

$6x - 17 = 7y$

$\frac{6}{7}x - \frac{17}{7} = y$

Always remember:

1. make y subject
2. Then compare it with $y = mx + c$

If a point lies on a line it must satisfy its equation



meaning
If point lies in line, put its coordinate in equation of line

3 Types of Question

① Check if (2, 5) lies on $y = 3x+2$ or not



$y = 3x + 2$

$5 = 3(2)+2$

$5 = 8$ NO

That means point does not lie on line

LHS ≠ RHS

② Given that (3, k) lies on line $3x+4y=12$, find value of k.

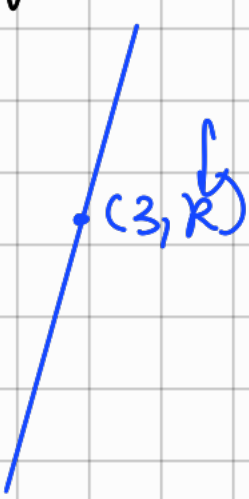
$3x + 4y = 12$

$3(3)+4(k)=12$

$9+4k=12$

$4k=3$

$k = \frac{3}{4}$



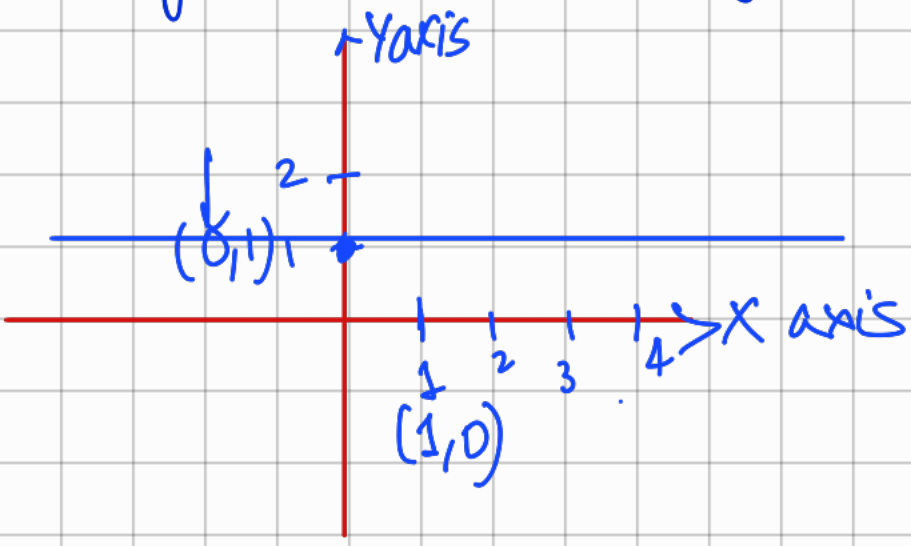
(3) Given that $2y = x - 3$ passes through point $(t, 5)$, find t.

$10 = t - 3$

$\boxed{t = 13}$

★ If a line cut x axis put $y = 0$ ✓

★ If a line cuts y axis put $X = 0$



coordinate geometry

(a) A is the point $(3, 7)$ and B is the point $(-1, 5)$.

Q Find midpt A (3,7) B (-1,5)

(i) Find the coordinates of the midpoint of the line AB.

Midpt formula $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$= \frac{3+(-1)}{2}, \frac{7+5}{2}$

$= \left(\frac{2}{2}, \frac{12}{2}\right)$

(..... 1 ..... , ... 6 .....) [2]

1. A is the point $(2, 3)$ and B is the point $(7, -5)$.

(a) Find the co-ordinates of the mid point of AB. ---[2]

(b) Find the equation of the line through A that is perpendicular to AB. Give your answer in the form $y = mx + c$. ---[4]

M-19/22/Q23

their gradient are negative reciprocal of each other

x_1 y_1 x_2 y_2

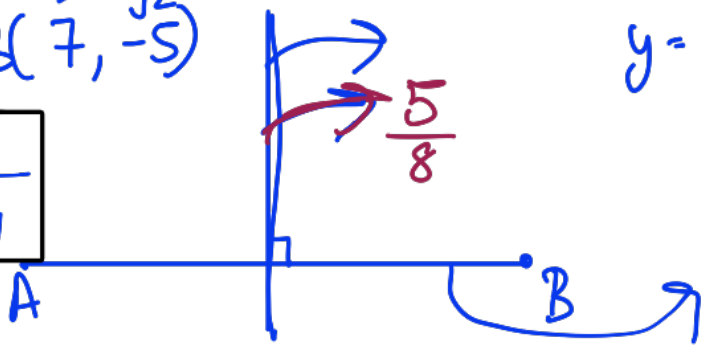
(a) A (2,3) , B (7, -5) midpt of AB → $\left(\frac{4}{2}, -1\right)$

(b) A (x_1 2, y_1 3) B (x_2 7, y_2 -5)

$y = mx + c$

gradient of AB $= \frac{y_2 - y_1}{x_2 - x_1}$

$\Rightarrow \frac{-5-3}{7-2} = \left(\frac{-8}{5}\right)$



gradient of line perpendicular to AB $= \frac{5}{8}$ ✓

Negative reciprocal

Find eq of line passes through A (2,3)

$y = mx + c$ ←

$\rightarrow \quad y - y_1 = m(x - x_1)$

$$y - 3 = \frac{5}{8}(x - 2)$$

$$y - 3 = \frac{5x}{8} - \frac{10}{8}$$

$$y = \frac{5x}{8} - \frac{10}{8} + \frac{3}{1}$$

$$= \frac{5x}{8} \quad \frac{-10+24}{8}$$

$$= \frac{5x}{8} + \frac{14}{8} \quad \begin{matrix} 7 \\ 4 \end{matrix}$$

eq of line which is $\perp$ to AB $= \frac{5x}{8} + \frac{7}{4}$ ✓

coordinate geometry

2. (a) Find the co-ordinates of the point where the line $y = 3x - 8$ (crosses the y-axis). --[1]

(b) Write the gradient of the line $y = 3x - 8$ [S-19|23|Q5] --[1]

(a) Put $x = 0$

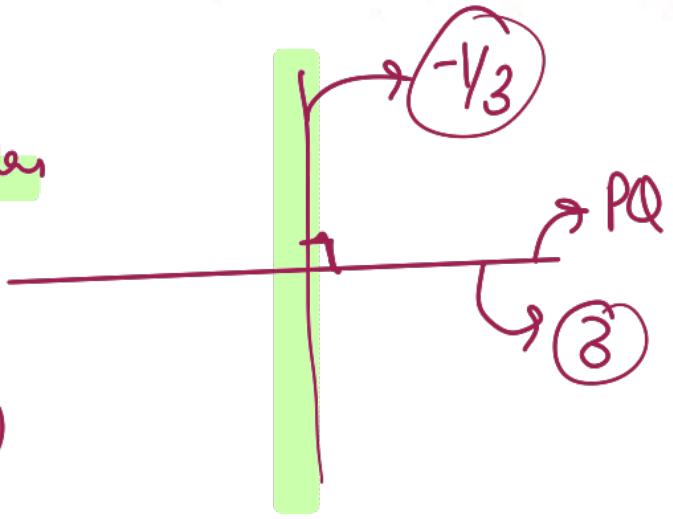
$y = 0 - 8$ $y = -8$ $(0, -8)$

(b) $m = 3$

(b) The line PQ has equation $y = 3x - 8$ and point P has co-ordinates $(6, 10)$. Find the equation of the line that passes through P and is perpendicular to PQ. Give your answer in the form $y = mx + c$. [S-18|21|Q24] --[3]

gradient of PQ = 3

gradient of line perpendicular to PQ $= -\frac{1}{3}$



$y - y_1 = m(x - x_1)$

$y - 10 = -\frac{1}{3}(x - 6)$

$y = -\frac{1}{3}x + \frac{16}{3}$

$\frac{6+10}{3} = \frac{16}{3}$

$$y - 10 = -\frac{1}{3}x + \frac{6}{3}$$

$$y = -\frac{1}{3}x + \frac{6}{3} + \frac{10}{1}$$

$$\boxed{y = -\frac{1}{3}x + 12}$$